A SESSION OF THE 101ST KPPY WORKSHOP

Young KPPY

DATE
November 13, 2026 (Friday)

Talks 14:00–17:30, dinner 18:00

VENUE
B06, College of Natural Sciences
Kyungpook National University, Daegu

youngkppy.seminresearch.com

PROGRAM

14:00–14:45	**Hong-Jun Ge** University of Science and Technology of China On co-edge-regular graphs with four distinct eigenvalues
14:45–14:55	Break
14:55–15:40	**Hyungkee Yoo** Sunchon National University On Three- and Four-Page Presentations of Knots and Links
15:40–15:50	Break
15:50–16:35	**Sunyo Moon** Kumoh National Institute of Technology Title to be announced
16:35–16:45	Break
16:45–17:30	**Jingyuan Liu** University of Science and Technology of China On signed graphs with smallest eigenvalue at least -3 and their lattices
18:00	Dinner

Each talk is 40 minutes plus 5 minutes for questions.

14:00–14:45

On co-edge-regular graphs with four distinct eigenvalues

Hong-Jun Ge University of Science and Technology of China

Joint work with Edwin R. van Dam and Jack H. Koolen

A connected k-regular graph on v vertices is called co-edge-regular with parameters (v, k, μ) if every pair of distinct nonadjacent vertices has exactly μ common neighbours. The λ-level of a graph G is the number of distinct values of $|N(x) \cap N(y)|$ as xy ranges over the edges of G.

Tan, Koolen, and Xia conjectured that, for a fixed smallest eigenvalue under suitable largeness and density assumptions, every connected co-edge-regular graph with exactly four distinct eigenvalues must be either a clique extension of a strongly regular graph or a grid graph. Both have λ-level 2.

In this talk, we present two constructions yielding infinite families of counterexamples. Every graph constructed is cospectral with a clique extension of a strongly regular graph, but has λ-level larger than 2.

The first construction starts from a regular partial linear space containing an affine or projective plane and applies Wang–Qiu–Hu (WQH) switching. For every order q for which such a plane exists, the construction yields infinitely many graphs with smallest eigenvalue $-2q - 1$ and λ-level at least $q + 2$. Thus, the λ-levels arising from this construction are unbounded as q varies.

The second construction is based on group-divisible orthogonal arrays and parallel classes arising from vector spaces over finite fields. The graphs in this family have smallest eigenvalue -5 and λ-level 3. More strikingly, the coherent closure of each graph is the full matrix algebra, so its coherent rank attains the maximum possible value v^2, where v denotes its number of vertices.

These constructions show that, even in the co-edge-regular setting, four eigenvalues does not preclude either unbounded local complexity or maximal coherent-algebraic complexity.

14:55–15:40

On Three- and Four-Page Presentations of Knots and Links

Hyungkee Yoo Sunchon National University

Page presentations provide combinatorial representations of knots and links by embedding them into a union of half-planes sharing a common boundary line. The corresponding page indices measure the minimum number of arcs required for such representations.

In this talk, I will introduce the history of three-page presentations and review some known results on the three-page index. I will then present a diagrammatic approach based on non-

separating independent sets in planar graphs, which yields the upper bound

$$\alpha_3(L) \leq 3c(L) - 1$$

for any non-split, nontrivial link L other than the Hopf link.

I will also introduce the four-page index and explain how spanning trees of checkerboard graphs lead to the improved bound

$$\alpha_4(L) \leq 2c(L).$$

Finally, I will discuss applications to ribbonlength and related combinatorial problems.

15:50–16:35

Title to be announced

Sunyo Moon Kumoh National Institute of Technology

Abstract to be announced.

16:45–17:30

On signed graphs with smallest eigenvalue at least -3 and their lattices

Jingyuan Liu University of Science and Technology of China

Joint work with Meng-Yue Cao, Jack H. Koolen, and Qianqian Yang

Graphs with smallest eigenvalue at least -2 can be represented by root lattices. In 2018, Koolen et al. gives a result on (unsigned) graphs with smallest eigenvalue at least -3 and their associated lattices. This talk presents an extension of this theory to signed graphs.

We prove that every connected signed graph with smallest eigenvalue greater than $-3-\varepsilon$ and sufficiently large minimum valency has smallest eigenvalue at least -3, and its associated lattice is a sublattice of $\mathbb{Z}^n \oplus E_8^m$ for some n, m.

Consequently, this also gives a result on rootless lattices generated by norm 3 vectors.